

Task 3: API Integration Project

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Role	Frontend Developer Intern
Task	Task 3 — API Integration Project
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Objective

Build a web application that fetches live data from a third-party API using JavaScript Fetch API. The Weather App demonstrates real-time API integration by retrieving current weather data for any city using the OpenWeatherMap API.

Technologies Used

Technology	Purpose
HTML5	Structure and layout of the application
CSS3	Styling, gradient background, responsive design
JavaScript ES6+	API calls using Fetch, DOM manipulation
OpenWeatherMap API	Real-time weather data source (free tier)
Vercel	Deployment and hosting

Key Features

- City-based weather search with real-time API fetch
- Displays temperature (°C), city name, humidity percentage, and wind speed
- Dynamic weather icons based on current conditions
- Error handling for invalid city names
- Responsive design — works on desktop and mobile
- Dark-themed UI with gradient card design

API Integration — Code Snippet

The app uses the JavaScript Fetch API to call the OpenWeatherMap endpoint:

```
const apiUrl =
`https://api.openweathermap.org/data/2.5/weather?units=metric&q=`;

async function checkWeather(city) {
  const response = await fetch(apiUrl + city + `&appid=${apiKey}`);
  if (response.status === 404) {
    document.querySelector(".error").style.display = "block";
  } else {
    const data = await response.json();
    document.querySelector(".temp").innerHTML =
Math.round(data.main.temp) + "°C";
    document.querySelector(".humidity").innerHTML = data.main.humidity +
"%";
    document.querySelector(".wind").innerHTML = data.wind.speed + "
km/h";
  }
}
```

Project Links

Live Demo:

<https://weather-app-lilac-pi-19.vercel.app/>

GitHub Repository:

<https://github.com/Daniish-Qureshi/Weather-App>

Task Folder (Alfido Tech Repo):

<https://github.com/Daniish-Qureshi/Alfido-Tech-Internship/tree/main/Task-3-API-Integration>

Screenshots

Initial State — Search Interface

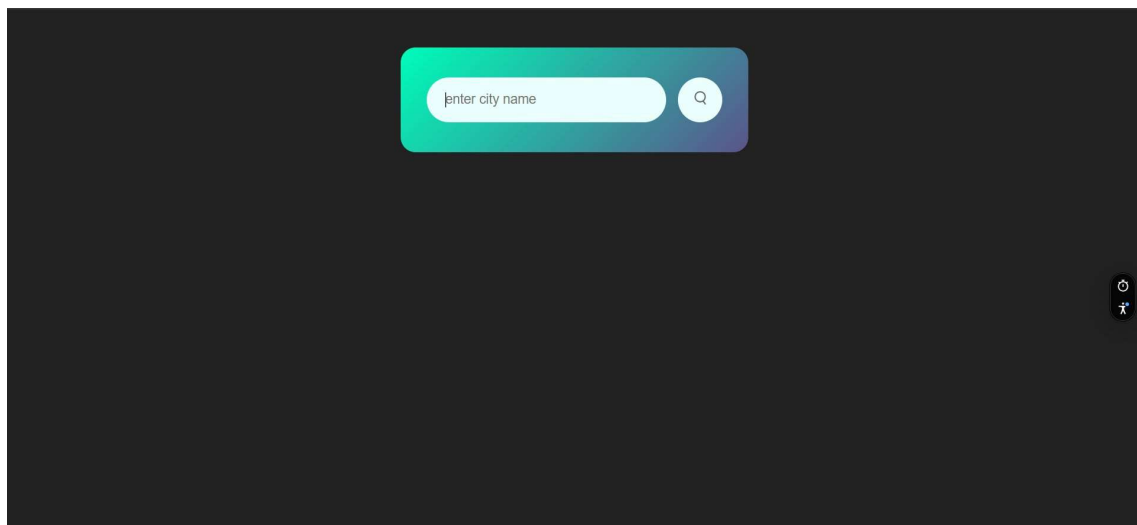


Figure 1: Initial state showing city search input with gradient card UI on dark background

Search Result — Dadri Weather Data

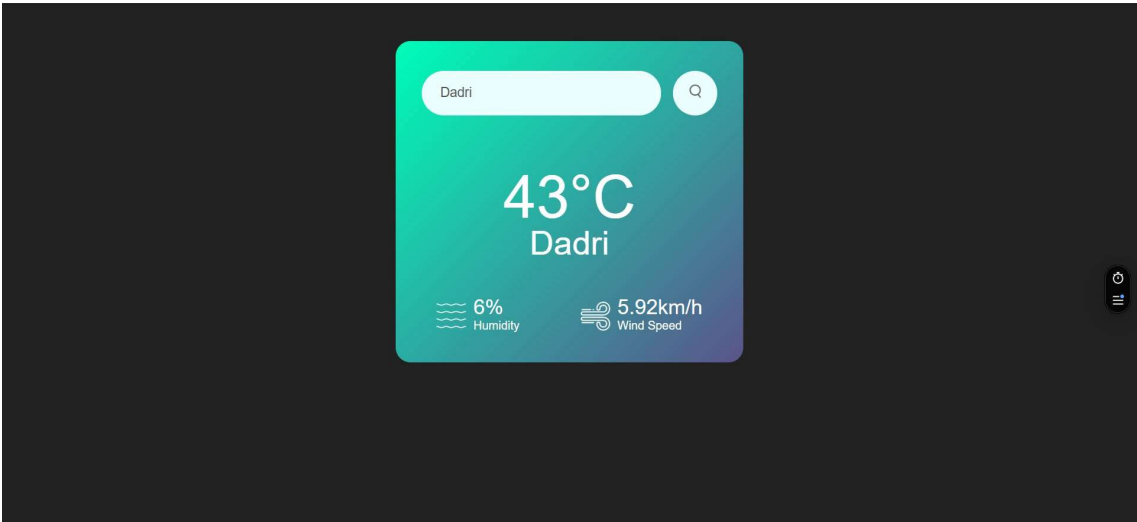


Figure 2: Weather result for Dadri — 43°C temperature, 6% humidity, 5.92 km/h wind speed fetched live from OpenWeatherMap API

Mobile View — Responsive Layout



Figure 3: Mobile responsive view showing weather data for Dadri on small screen

Implementation Summary

The Weather App successfully demonstrates API integration using vanilla JavaScript. On city search, the app makes an asynchronous Fetch request to the OpenWeatherMap API endpoint, parses the JSON response, and dynamically updates the DOM with temperature, humidity, and wind speed data. Invalid city names are handled gracefully with an error message. The app is fully responsive and deployed live on Vercel.

Submitted by Danish Qureshi | Alfido Tech Frontend Developer Internship | May 2026